Physical Geology
G 101

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مقرر علم الجيولوجيا الطبيعية (G 101)
Physical Geology

أولاً / مخطط المقرر:

اسم المادة	جيولوجيا طبيعية
رمز المادة ورقمها	**G 101**
المتطلبات السابقة	-
القسم	جيولوجيا
السنة الدراسية	الفصل الدراسي الاول 2020 م
أسا تذة الجزء النظرى	ا.د. رافت ذكى د. احمد جمعة
مواعيد	يوم السبت 11-12 ص يوم الاثنين 11-12 ص يوم الاربعاء 10-11 ص

ثانياً / الأهداف العامة للمقرر:

1- Overall Aims of Course

.Know the Earth history and origin.

.Understand the external and internal processes that affect on the earth crust. .Understand the orogenic and epiorogenic movements.

.Understand the geologic activity of biosphere

2- Intended Learning outcomes of course (ILOs)

a- Knowledge and Understanding

a1. Recognize the basic concepts of geological sciences such as structures and petrology, mineralogy, and economic geology.

a2. Understand the evolution of the geological history of the Earth.

a3. Acquire an understanding of economic evaluation of minerals, rocks, and of the Earth.

b- Intellectual skills

b1. Distinguish between the different rocks

b2. interpret the structural and subsurface and surface geology in exploration of oil, gas, water and economic minerals and elements.

b3. Predict and solve the geological geo-environmental problems.

d- General and Transferable Skills

d1. Solve the geological problems using the suitable methods.

d2. Present result of research using written and oral means.

d3. Work effectively both in team, and independently on geological research and in industry.

d4. Share team work in the field work in order to perform the geologic maps.

d5.Management of self time, data and knowledge in solution of the geological problems.

3- Contents

TOPIC	No. OF HOURS	LECTURE	TUTORIAL/PRACTICAL
Introduction	3	1	2
Solar System and the Theory		1	2
Plate tectonic theory		1	2
primary structures		1	2
Secondary structures		1	2
Minerals		1	2
Mid-term Exam			
Sedimentary rocks		1	2
Igneous rocks metamorphic rocks		1	2
Weathering		1	2
Earthquakes		1	2
Groundwater		1	2
Volcanoes		1	2

رابعاً/ المراجع الخاصة بالمقرر:

6-3- Recommended books

*Physical Geology : Answers.com

www.answers.com/topic/igneous-rock

* Physical Geology (Pictures of Intrusive and Extrusive Rock types

www.answers.com/search?q=physical-geology

*Introduction to Physical Geology

http://forum.arabsbook.com/threads/introduction-to-physical-geology.37819/#ixzz1qVF59Mkm

** *Physical Geology*: The Science of Earth *www.wiley.com/.../productCd-EHEP001789.html*, Wiley:
*Laboratory Manual in Physical Geology by AGI - American Geological Institute (9th Edition)

*Introduction to Physical Geology http://iogeophysicist.blogspot.com/2012/02/introduction
*Hundman , D.W. (1972: Petrology of Igneous and Metamorphic rocks. McLerw. Ital, New York.

What is the Geology?

*The word means "study of the Earth". Geology deals with the composition of Earth materials, Earth structures, and Earth processes.

*It is also concerned with the organisms of the planet and how the planet has changed over time.

*Geologists search for fuels and minerals, study natural hazards, and work to protect Earth's environment.

What do geologists do?

Geology is a multi-faceted field with many different areas of specialization. ***Earth Science:*** A mixture of geology, oceanography and climatology) in junior and senior high schools. A teaching certificate from a professional education program is also normally required.

2-***Economic Geology:***

explore for and help produce metallic (iron, copper, gold, etc.) and non-metallic (coal, granite dimension stone, limestone aggregate, sand and gravel, etc.) rock and mineral resources of economic value.

3-Engineering Geology:

investigate the engineering properties of rock, sediment and soil below man-made structures such as roads, bridges, high-rise buildings, dams, airports, etc.

4-*Environmental Geology:*

study the environmental effects of pollution on ground and surface waters and surficial materials (rock, sediment and soil), and also recommend solutions to environmental problems. They are also interested in understanding, predicting and mitigating the effects of natural hazards, such as flooding, erosion, landslides, volcanic eruptions, earthquakes, etc.

5-***Geochemistry:***

investigate the chemical composition and properties of earth materials, especially polluted ground and surface waters, fossil fuels (such as petroleum and coal) and other resources of economic value.

6-***Geomorphology:***

study the origin and evolution of landscapes on the continental surfaces.

7-***Geophysics:***

use the principles of physics to investigate the structure of the Earth's deep interior, explore for economic resources in the subsurface, and monitor pollution in ground water.

8-***Glacial or Quaternary Geology:***

study the history of geologically recent (Quaternary period) glaciers as well as the sediment deposits and landforms they produced.

9-Hydrogeology:

are concerned with water in the Earth's subsurface, including its sources, quality, abundance and movement.

10-***Hydrology:***

are concerned with water on the Earth's surface, including its precipitation, evaporation and runoff, and its abundance and quality in streams and lakes.

11-***Marine Geology:***

study the physical, chemical and biological characteristics of the sediments deposited on the ocean floors and the rocks that underlie them.

12-***Mineralogy:***

investigate the origins, properties and uses of the minerals occurring within the Earth's rocks.

13-***Paleontology:***

study the remains of ancient animals and plants (fossils) in order to understand their behaviors, environmental circumstances, and evolutionary history.

14-***Petroleum Geology:***

explore for and help produce petroleum and natural gas from sedimentary rocks.

15-***Petrology:***

study the origins and characteristics of igneous, metamorphic and sedimentary rocks.

16-***Sedimentology:***

investigate the origins and characteristics of sediment deposits and the sedimentary rocks that form from them.

17-***Stratigraphy:***

investigate the time and space relationships among sedimentary and other rocks on local to global scales, and are also interested in the geochronology (absolute dating by radiometric methods) and fossil content of rock layers.

18-***Structural Geology:***

study the folding, fracturing, faulting and other forms of deformation experienced by rocks below the Earth's surface,

and are also interested in how these processes relate to global Plate Tectonics.

19-Physical Geology:

discuss the processes that affect the earth crust as weathering, surface water and groundwater ….etc as well as volcanoes and earthquakes.

20 **-Historical Geology:**

Examines origin of the Earth, origin of life, and changes in Earth and life through time. It deals with the layered rock record and fossils.